

This is feasible in four months, but the project should be divided into two levels:

1. **One-way photon-pair clock-offset estimation over a calibrated, fixed fiber** using your present source and two time taggers.
2. **Round-trip compensation**, which separates the clock offset from changes in fiber propagation delay.

The second level is what makes the experiment a convincing synchronization system rather than only a fiber-delay measurement.

1. The fundamental measurement

Let the source generate a pair at an unknown physical time t_0 . One photon is detected at Alice and the other at Bob:

$$t_A = t_0 + d_A + C_A,$$
$$t_B = t_0 + T_{AB} + d_B + C_B + \Delta,$$

where:

- T_{AB} : fiber propagation time;
- Δ : Bob-clock offset relative to Alice;
- d_A, d_B : optical and detector delays;
- C_A, C_B : time-tagger and cable delays.

The measured coincidence position is

$$z = t_B - t_A = T_{AB} + \Delta + C,$$

with $C = d_B - d_A + C_B - C_A$.

This immediately reveals an important limitation:

One-way measurement cannot distinguish clock offset from fiber delay.

A pre-calibration removes the initial constant:

$$z - z_{\text{ref}} = [\Delta - \Delta_{\text{ref}}] + [T_{AB} - T_{AB,\text{ref}}].$$

Therefore, the one-way method works when:

- the fiber remains fixed and stable;
- you deliberately vary only the clock offset;
- or you are measuring the combined timing change.

It does not provide absolute synchronization if the fiber delay is independently changing.

2. Recommended project claim

A realistic and strong four-month objective would be:

Demonstrate photon-pair-based clock-offset estimation between independent time-tagging modules, followed by round-trip cancellation of fiber propagation-delay fluctuations, with a target repeatability of 10 ps.

Recent experiments have demonstrated clock-offset extraction from independent timestamp modules using commercial photon-pair hardware, and direct cross-correlation methods have reported sub-picosecond statistical peak-position precision under favorable count rates.  [arXiv +1](#)

3. Hardware you need

Already available

- Si_3N_4 SFWM photon-pair source
- pump laser and pump suppression
- signal/idler filters or DWDM
- two time taggers
- single-photon detectors
- fiber coupling and polarization control

Essential additions

Item	Purpose
Two independent clock references	Establish genuinely separate Alice and Bob timescales
10 MHz and preferably 1 PPS outputs	Feed the external-reference inputs of the time taggers
Programmable electronic delay generator	Apply known clock phase offsets for calibration
Variable optical delay or calibrated fiber lengths	Test known physical delays
Fiber spools	0–10 km laboratory link
Temperature sensors	Record fiber, source, detector and tagger temperatures
Optical attenuator	Reproduce link loss independently of fiber length

Item	Purpose
Classical Ethernet connection	Transfer timestamp files; it does not need picosecond synchronization

For the clock references, rubidium, chip-scale atomic clocks, good OCXOs or GPS-disciplined oscillators can be used. The exact choice controls how quickly the clocks drift apart. If your taggers only use their own internal oscillators, you can begin with those, but their frequency difference must be estimated and corrected.

Round-trip additions

- optical circulators;
- a 50:50 coupler at Bob;
- one additional server-side detector channel;
- ideally three detector channels in total:
 - Alice reference photon;
 - Bob outbound photon;
 - Alice returned photon.

Having two time taggers is sufficient if the Alice tagger has at least two detector inputs and the Bob tagger has at least one.

If only two detector channels are available, outbound and round-trip measurements can be time-multiplexed with an optical switch, but this is less convincing because fiber delay may change between the two measurements.

Experimental program

Experiment 1: Time-tagger electrical calibration

Before using photons, characterize the taggers themselves.

Setup

Split the same electrical pulse into both time taggers:

```

Pulse generator
  |
RF splitter
 /   \
Tagger A   Tagger B
Clock A   Clock B

```

Start with both taggers connected to the same 10 MHz reference.

Measurements

Measure:

- fixed channel-to-channel delay;
- histogram width;
- fine-bin ripple;
- peak-center stability;
- delay versus input count rate;
- delay versus temperature;
- delay versus cable arrangement;
- delay versus discriminator threshold, if adjustable.

Insert known electronic delay changes:

0, 10, 20, 50, 100, 500 ps.

Randomize the order so that slow drift is not mistaken for linearity.

Output

For each applied delay D_k , obtain an estimate $\hat{D}_k$ and fit

$$\hat{D}_k = aD_k + b.$$

Target:

$$a \approx 1, \quad |\hat{D}_k - D_k| < 5\text{--}10 \text{ ps.}$$

This determines whether the electronics alone can support your target.

Experiment 2: Full photon-pair source calibration

Setup

```
Pump
|
SiN spiral source
|
Signal/idler demultiplexer
|           |
Detector A   Detector B
|           |
Tagger A     Tagger B
shared clock
```

Data to collect

For every run, save the raw timestamps rather than only the histogram.

Measure as a function of pump power:

- singles rates S_A, S_B ;
- coincidence rate;
- accidental rate;
- CAR;
- fitted peak center;
- peak RMS width and FWHM;
- fit uncertainty;
- background level;
- source spectrum and filter bandwidth;
- detector count rates;
- room and device temperatures.

Also repeat the measurement at several filter bandwidths. A recent independent-clock experiment reported that spectral filtering improved timing accuracy, although narrowing the filter too much will reduce count rate and can increase the biphoton coherence time.

 Optica Publishi...

Source stability run

Run the source continuously for at least 6–12 hours and calculate the peak center in consecutive blocks.

Use blocks defined by both:

- fixed duration, such as 1, 5, 10 and 30 seconds;
- fixed true-coincidence count, such as 500, 1,000 and 10,000 pairs.

Plot:

$$\hat{\tau}(t), \quad \sigma_{\text{peak}}(t), \quad S_A(t), \quad S_B(t), \quad \text{CAR}(t).$$

This is particularly important because SNSPD timing can depend on count rate, and experiments have observed changes in coincidence-peak position when photon count rates vary.  ar...

Experiment 3: Validate 10 ps delay estimation

This experiment establishes whether your complete optical system actually resolves a controlled 10 ps change.

Optical implementation

A delay of 10 ps corresponds to approximately:

$$\Delta L_{\text{fiber}} = \frac{c\Delta t}{n_g} \approx 2.0 \text{ mm}$$

of one-way optical-fiber length for $n_g \approx 1.47$.

A free-space retroreflector is more convenient:

$$\Delta t = \frac{2\Delta x}{c}.$$

Thus, a retroreflector displacement of approximately

$$\Delta x = 1.5 \text{ mm}$$

introduces about 10 ps of round-trip optical delay.

A calibrated RF delay generator on a detector output is even easier for validating the estimator, although it does not test the optical path.

Delay sequence

Apply:

$$0, \pm 5, \pm 10, \pm 20, \pm 50, \pm 100 \text{ ps}.$$

Repeat every setting at least 10–20 times.

Analysis

Use a two-stage estimator:

1. Coarse cross-correlation with relatively large bins to find the peak.
2. Fine analysis around the peak using 10–20 ps bins and template fitting.

Do not use the maximum-bin position.

Fit the shifted reference template:

$$H_k(t) = A_k H_{\text{ref}}(t - \tau_k) + B_k.$$

Test the estimator with:

- 5 ps bins;
- 10 ps bins;
- 20 ps bins;
- several bin origins;
- unfiltered data;

- symmetrically filtered data.

The recovered delay should remain stable within approximately 1–2 ps under these analysis choices if you want to claim 10 ps system precision confidently.

Count requirement

For a peak with RMS width σ_t ,

$$\sigma_{\hat{\tau}} \approx \frac{\sigma_t}{\sqrt{N_{\text{eff}}}}.$$

If your 200 ps width is RMS,

$$N_{\text{eff}} \gtrsim \left(\frac{200}{10} \right)^2 = 400$$

is the ideal statistical minimum for 10 ps precision.

However, background, accidentals, clock drift and model mismatch make this optimistic. I would collect at least

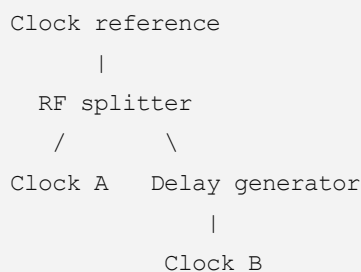
10^4 true coincidences per calibration condition

and then separately determine how few counts are needed for real-time estimates.

Experiment 4: One-way clock-offset estimation

Stage A: Artificial clock offset

Use one clock reference split into two paths, with a programmable delay in Bob's clock input:



Now the applied clock offset is known independently.

Measure the photon-pair coincidence peak while changing the programmed offset. This tests the exact operation you ultimately want:

$$\Delta_{\text{estimated}} = \tau_{\text{measured}} - \tau_{\text{reference}}.$$

This stage should demonstrate:

- offset linearity;
- offset accuracy;
- repeatability;
- acquisition-time versus precision;
- robustness against photon loss.

Stage B: Independent clocks

Replace the shared reference with separate oscillators:

Clock A → Tagger A

Clock B → Tagger B

Now the clock relationship is not only an offset. It is approximately

$$t_B = (1 + \epsilon)t_A + \Delta,$$

where:

- Δ : phase offset;
- ϵ : fractional frequency difference.

Consequently, the coincidence peak drifts:

$$\tau_{\text{peak}}(t) = \tau_0 + \epsilon t.$$

Estimate both the intercept and slope:

$$\hat{\Delta} = \tau_0, \quad \hat{\epsilon} = \frac{d\tau_{\text{peak}}}{dt}.$$

Photon-correlation experiments have used correlation-peak motion to recover both clock offset and frequency difference between independent clocks. [arXiv +1](#)

Block length

Without frequency correction, require

$$|\epsilon|T_{\text{block}} \ll \sigma_{\text{peak}},$$

and preferably

$$|\epsilon|T_{\text{block}} < 10 \text{ ps}$$

for a 10 ps target.

Otherwise, the moving peak is broadened during acquisition. You can either shorten the blocks or apply an estimated frequency-skew correction before constructing the final

histogram.

Experiment 5: Fiber propagation study

Insert successively:

- short patch cable;
- 1 km;
- 5 km;
- 10 km fiber spools.

For each length collect:

- total propagation delay;
- optical loss;
- coincidence rate;
- CAR;
- peak width;
- peak asymmetry;
- estimated clock offset;
- fiber temperature;
- polarization state or controller setting.

The fiber introduces both attenuation and chromatic-dispersion broadening. Metropolitan timing experiments have also identified path-delay gradients, dispersion, polarization changes and optical-power variations as significant synchronization-error mechanisms.

 NIST

Temperature test

Standard solid-core telecom fiber has a propagation-delay temperature coefficient around

$$30\text{--}40 \text{ ps}/(\text{km} \cdot \text{K}).$$

Thus, for 10 km of fiber, a 1 K change can move the one-way delay by roughly

$$300\text{--}400 \text{ ps},$$

far larger than your 10 ps target.  IEEE Xplore +1

This makes an excellent controlled experiment:

1. Place part or all of the fiber spool in an insulated box.
2. Change its temperature slowly by 1–3 K.
3. Monitor the one-way coincidence peak.
4. Demonstrate that the apparent clock offset follows the fiber temperature.

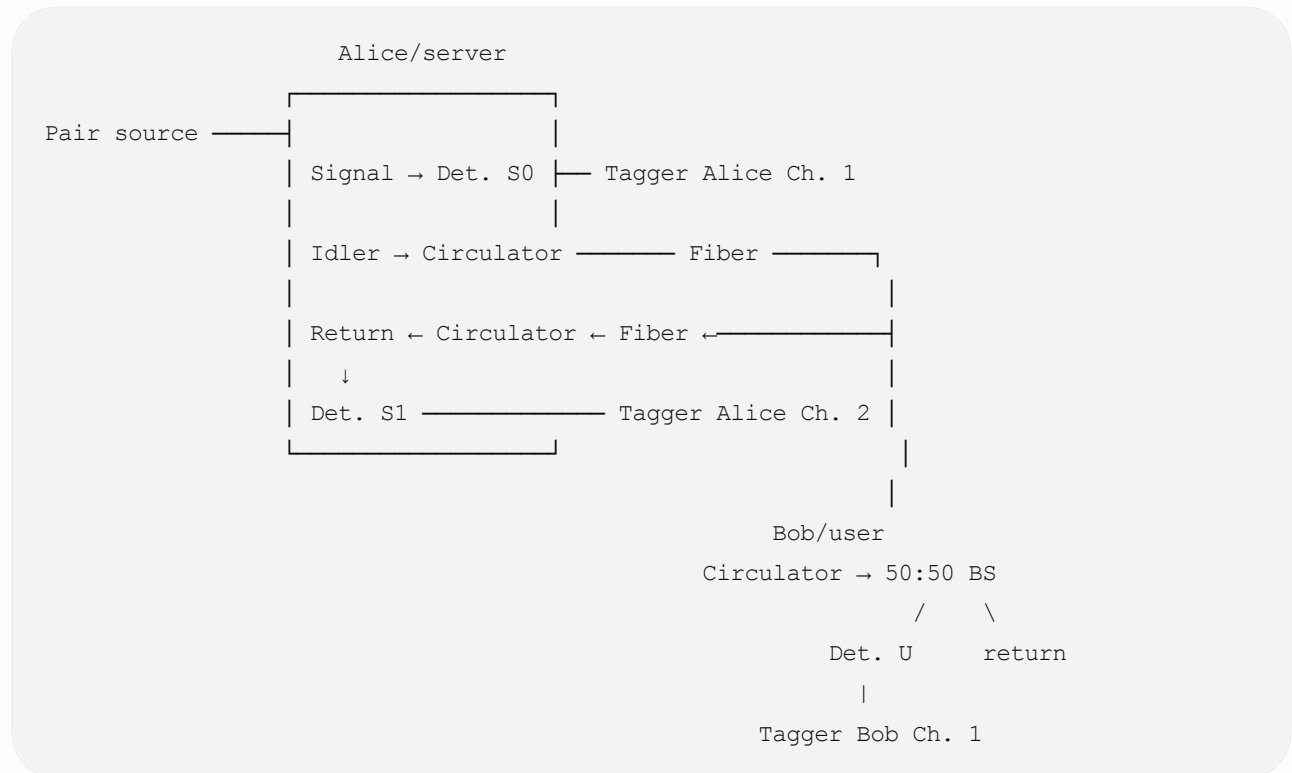
5. Then demonstrate cancellation using the round-trip method.

4. Round-trip synchronization setup

This is the most valuable final experiment.

Setup

Place the source at Alice/server:



At Bob, one portion of the arriving idler is detected locally and the other is returned through the same fiber.

A recent chip-source QCS network used this type of signal-at-server, idler-to-user and partial-round-trip architecture, with local TDCs referenced by their respective clocks.

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Timing equations

Let the Alice signal photon provide the reference timestamp s .

The Alice-to-Bob measurement is

$$x = t_U - t_{S0} = T + \Delta + C_1.$$

The round-trip measurement is

$$y = t_{S1} - t_{S0} = 2T + C_2.$$

Therefore,

$$\hat{\Delta} = x - \frac{y}{2} - C_{\text{cal}}$$

where

$$C_{\text{cal}} = C_1 - \frac{C_2}{2}$$

is measured during co-located or zero-length calibration.

The fiber delay T cancels.

Calibration requirements

The cancellation assumes:

- forward and return propagation are reciprocal;
- the same fiber path is used;
- preferably the same wavelength is used outbound and return;
- component delays remain stable;
- detector and tagger offsets are calibrated.

Residual errors come from:

- circulator and coupler asymmetry;
- wavelength-dependent group delay;
- polarization-dependent delay;
- detector count-rate-dependent timing;
- different detector and cable delays;
- temperature differences between Alice and Bob.

A chip-based round-trip experiment obtained picosecond-level TDEV but still observed nonzero residual offsets attributed to component asymmetry and SNSPD timing drift.

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5. Data products for every run

Use one metadata record for each acquisition:

```
run_id
date and start time
source type
pump wavelength and power
signal/idler wavelengths
filter bandwidths
fiber length
```

```
fiber temperature
room temperature
detector identities
detector biases
singles rates
tagger identities and channel mapping
clock-reference identities
applied clock delay
applied frequency offset
raw timestamp filenames
analysis-code version
```

From every timestamp block extract:

$\hat{\tau}$	peak center,
σ_{τ}	fit uncertainty,
σ_{peak}	peak width,
A	true-coincidence amplitude,
B	background,
CAR	coincidence-to-accidental ratio.

For the final performance figures, report:

- accuracy against known clock offsets;
- repeated-measurement standard deviation;
- delay linearity;
- frequency-skew estimation error;
- precision versus acquisition time;
- precision versus coincidence count;
- offset versus fiber temperature;
- one-way versus round-trip results;
- TDEV versus averaging time.

TDEV is widely used for characterizing time-transfer stability, including recent metropolitan and photon-pair synchronization demonstrations. [arXiv +1](#)

6. Four-month schedule

Weeks	Main objective	Deliverable
1–2	Electrical tagger calibration	Fixed delays, drift and fine-bin artifact characterization

Weeks	Main objective	Deliverable
3–4	Photon-source characterization	Stable reference histogram and source operating point
5–6	Known-delay recovery	Demonstrate recovery of 5–100 ps programmed delays
7–8	One-way common-clock experiment	Recover artificial clock phase offsets
9–10	Independent-clock operation	Estimate phase offset and frequency skew
11–12	Fiber length and temperature tests	Quantify loss, dispersion and thermal delay
13–14	Round-trip setup	Demonstrate fiber-delay cancellation
15	Long-duration measurements	6–12 hour runs and TDEV
16	Final analysis and presentation	Figures, uncertainty budget and demonstration

Minimum successful outcome

By approximately week 8, you should have:

Photon-pair-based recovery of known clock offsets with a fixed calibrated optical path.

Strong final outcome

By week 14–16:

Clock-offset estimation between independent clocks, with round-trip cancellation of controlled fiber-delay changes.

Stretch outcome

A feedback loop that adjusts Bob’s clock phase or applies real-time timestamp correction:

$$\Delta_{\text{correction}}(t) = -\hat{\Delta}(t).$$

Without this feedback, the experiment is most accurately described as **clock-offset estimation** or **virtual synchronization**. With the correction applied to the clock or timestamp stream, you can describe it as an operational synchronization system.

Most important risk

The greatest risk is not the 200 ps coincidence width. With enough counts, that averages down. The primary risks are:

clock frequency drift + fiber-delay drift + detector/tagger systematic delay
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The project should therefore emphasize controlled delay injection, repeated calibration and round-trip cancellation—not only obtaining a narrow coincidence peak.